Syllabus

1. The bootstrap
   - A motivating example
   - The general procedure
   - About the implementation in R
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   - Classification trees
   - Bagging of classification trees
   - How much do you gain?
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3. Random forests (an improved bagging of trees)
   - What can be improved?
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Introduction

The bootstrap has other uses than those described above.
In particular, it allows us to design ensemble methods in statistical learning.

Bagging (Bootstrap Aggregating), which is the most famous approach in this direction, can be applied to both regression and classification.

Below, we mainly focus on bagging of classification trees, but it should be clear that bagging of regression trees can be performed similarly.

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The classification problem

In classification, one observes (X_i, Y_i), $i = 1, \ldots, n$, where

- X_i collects the values of K predictors on individual i, and
- Y_i is the class $(\in \{1, 2, \ldots, K\})$ to which individual i belongs.

The problem is to classify a new observation for which we only see x, that is, to bet on the corresponding value $y(\in \{1, 2, \ldots, K\})$.

A classifier is a mapping

$$\begin{aligned} \phi_{\mathcal{S}} : \mathcal{X} &\to \{1, 2, \ldots, K\} \\ x &\mapsto \phi_{\mathcal{S}}(x), \end{aligned}$$

that is designed on the basis the sample $\mathcal{S} = \{(X_i, Y_i),\ i = 1, \ldots, n\}$.

The classification problem

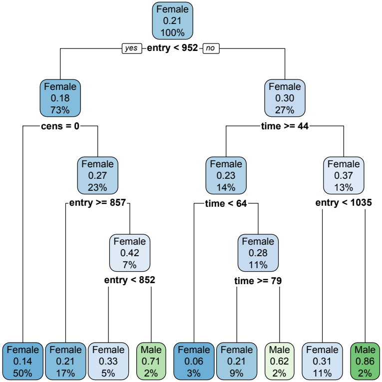
```
library(boot)
data(channing)
channing <- channing[,c("sex","entry","time","cens")]
channing[1:4,]

##    sex entry time cens
## 1 Male   782  127    1
## 2 Male  1020  108    1
## 3 Male   856  113    1
## 4 Male   915   42    1
```

Predict `sex` ($\in$ {Male, Female}) on the basis of two numerical predictors (`entry`, `time`) and a binary one (`cens`).

Classification trees

```
library(rpart)
library(rpart.plot)
fitted.tree <- rpart(sex~., data=channing, method="class")
rpart.plot(fitted.tree)
```



(+) Interpretability
(+) Flexibility
(–) Stability
(–) Performance

Classification trees

The process of averaging will reduce variability, hence, improve stability. Recall indeed that, if $U_1, \ldots, U_n$ are uncorrelated with variance σ^2, then

$$\mathrm{Var}[\bar{U}] = \frac{\sigma^2}{n}.$$

Since unpruned trees have low bias (but high variance), this reduced variance will lead to a low value of

$$\mathrm{MSE} = \mathrm{Var} + (\mathrm{Bias})^2,$$

which will ensure a good performance.

How to perform this averaging?

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Bagging

Denote as $\phi_{\mathcal{S}}(x)$ the predicted class for predictor value x returned by the classification tree associated with sample $\mathcal{S} = \{(X_i, Y_i),\ i = 1, \ldots, n\}$.

Bagging of this tree considers predictions from B bootstrap samples

$$\mathcal{S}^{*1} = ((X_1^{*1}, Y_1^{*1}), \ldots, (X_n^{*1}, Y_n^{*1})) \rightsquigarrow \phi_{\mathcal{S}^{*1}}(x)$$

$$\vdots$$

$$\mathcal{S}^{*b} = ((X_1^{*b}, Y_1^{*b}), \ldots, (X_n^{*b}, Y_n^{*b})) \rightsquigarrow \phi_{\mathcal{S}^{*b}}(x)$$

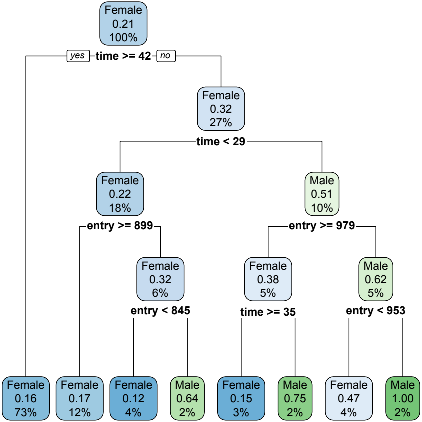
$$\vdots$$

$\mathcal{S}^{*B} = ((X_1^{*B}, Y_1^{*B}), \ldots, (X_n^{*B}, Y_n^{*B})) \rightsquigarrow \phi_{\mathcal{S}^{*B}}(x)$, then proceeds by majority voting (i.e., the most frequently predicted class wins):

$$\phi_{\mathcal{S}}^{\mathrm{Bagging}}(x) = \underset{k \in \{1,\ldots,K\}}{\arg\max}\ \#\{b : \phi_{\mathcal{S}^{*b}}(x) = k\}.$$

Toy illustration: bagging with $B = 3$ trees

```
d=sample(1:n,n,replace=TRUE)
fitted.tree <- rpart(sex~.,data=channing[d,],method="class")
rpart.plot(fitted.tree)
predict(fitted.tree, channing[1,], type="class")
```



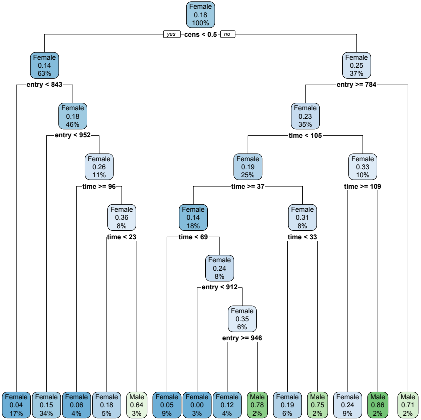
entry=782
time=127
cens=1

$\Downarrow$

Female

Toy illustration: bagging with $B = 3$ trees

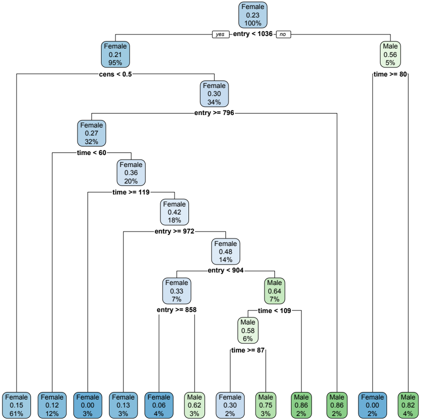
```
d=sample(1:n,n,replace=TRUE)
fitted.tree <- rpart(sex~.,data=channing[d,],method="class")
rpart.plot(fitted.tree)
predict(fitted.tree, channing[1,], type="class")
```



entry=782
time=127
cens=1
$\Downarrow$
Male

Toy illustration: bagging with $B = 3$ trees

```
d=sample(1:n,n,replace=TRUE)
fitted.tree <- rpart(sex~.,data=channing[d,],method="class")
rpart.plot(fitted.tree)
predict(fitted.tree, channing[1,], type="class")
```



entry=782
time=127
cens=1
$\Downarrow$
Male

Toy illustration: bagging with $B = 3$ trees

For $x = (\text{entry,time,cens}) = (782{,}127{,}1)$,

- two (out of the $B = 3$ trees) voted for Male
- one (out of the $B = 3$ trees) voted for Female,

the bagging classifier will thus classify x into Male.

Of course, B is usually much larger ($B = 500$? $B = 1000$?), which requires automating the process (through, e.g., the `boot` function).

(See the practical sessions).

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A simulation

To explore this, we performed the following simulation

We repeated $M = 1000$ times the following experiment:

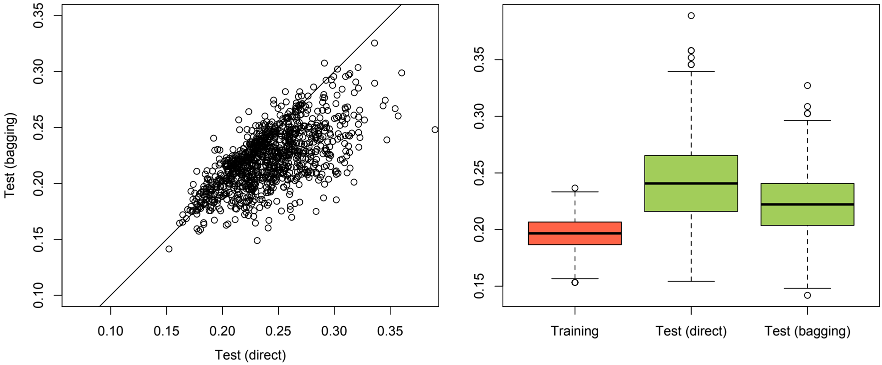
(1) Split the data set into a training set (of size 300) and a test set (of size 162);

(2a) train a classification tree on the training set and evaluate its test error (i.e., misclassification rate) on the test set;

(2b) do the same with a bagging classifier using $B = 500$ trees.

This provided $M = 1000$ test errors for the direct (single-tree) approach, and $M = 1000$ test errors for the bagging approach.

A simulation



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Estimating the prediction (lack of) accuracy

Several strategies to estimate prediction accuracy of a classifier:

(1) Compute a test error (as above):
Partition the data set $\mathcal{S}$ into a training set $\mathcal{S}_{\text{train}}$ (to train the classifier) and a test set $\mathcal{S}_{\text{test}}$ (on which to evaluate the misclassification rate e_{test}).

Estimating the prediction (lack of) accuracy

(2) Compute an L-fold cross-validation error:
Partition the data set $\mathcal{S}$ into L folds $\mathcal{S}_\ell$, $\ell = 1, \ldots, L$. For each ℓ, evaluate the test error $e_{\text{test},\ell}$ associated with training set $\mathcal{S} \setminus \mathcal{S}_\ell$ and test set $\mathcal{S}_\ell$.

Run ℓ=1	Test	Train	Train	Train	Train
Run ℓ=2	Train	Test	Train	Train	Train
Run ℓ=3	Train	Train	Test	Train	Train
Run ℓ=4	Train	Train	Train	Test	Train
Run ℓ=5	Train	Train	Train	Train	Test

The quantity

$$e_{\text{CV}} = \frac{1}{L} \sum_{\ell=1}^{L} e_{\text{test},\ell}$$

is then the (L-fold) "cross-validation error".

Estimating the prediction (lack of) accuracy

(3) Compute the Out-Of-Bag (OOB) error[1]:
For each observation X_i from $\mathcal{S}$, define the OOB prediction as

$$\phi_{\mathcal{S}}^{\mathrm{OOB}}(X_i) = \underset{k\in\{1,\ldots,K\}}{\arg\max} \, \#\{b : \phi_{\mathcal{S}^{*b}}(X_i) = k \text{ and } (X_i, Y_i) \notin \mathcal{S}^{*b}\}.$$

This is a majority voting discarding, quite naturally, bootstrap samples that use (X_i, Y_i) to train the classification tree.

The OOB error is then the corresponding misclassification rate

$$e_{\mathrm{OOB}} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}[\phi_{\mathcal{S}}^{\mathrm{OOB}}(X_i) \neq Y_i].$$

[1] This is for bagging procedures only.

Final remarks

Bagging of trees can also be used for regression.
The only difference is that majority voting is then replaced with an averaging of individual predicted responses.

Bagging is a general device that applies to other types of classifiers.
In particular, it can be applied to kNN classifiers
(we will illustrate this in the practical sessions).

Bagging affects interpretability of classification trees.
There are, however, solutions that intend to measure importance of the various predictors (see the next section).